

Real Time Estimation of GHG Emissions using IoT Integrated Sensor Fusion

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Abstract— IoT integrated Sensor Fusion is utilised in this work to estimate greenhouse gas (GHG) emissions. A system is proposed that utilizes IoT devices and environmental sensors to monitor and manage environmental impact efficiently in real-time. IoT-enabled sensors are strategically placed in industrial sites, metropolitan centres, and natural landscapes. These sensors measure CO₂, methane, nitrous oxide, and meteorology. Sensor fusion and real time data monitoring are the main contribution of this research work. The technology decreases uncertainty and improves estimation accuracy by integrating sensor data. Machine learning algorithms analyse fused sensor data and create predictive models. These models allow the system to extrapolate emission estimates beyond sensor locations, providing a comprehensive view of GHG emissions across geographies. IoT integration with sensor fusion and digital signal processing (DSP) allows real-time data transmission and remote monitoring. This continuously collects and processes emission data to quickly identify pollution hotspots and conduct emission reduction actions. The system's precision, real-time capabilities, scalability, and adaptability make it ideal for local air quality monitoring, regional emission tracking, and global climate change assessments.

Finally, this research uses sensor fusion and IoT technology to estimate GHG emissions. The technology provides accurate, real-time emission estimations from several sensors, aiding environmental monitoring and sustainable development. This strategy could help mitigate climate change and create a green future.

Keywordst; Greenhouse Gas (GHG), DSP, Sensor Fusion, IoT

I. INTRODUCTION

The unprecedented growth in industrialization, urbanization, and transportation has led to a significant increase in greenhouse gas (GHG) emissions, posing a grave threat to the planet's climate and ecosystem. Accurate and timely monitoring of GHG emissions is paramount for formulating effective climate change mitigation strategies and meeting international climate targets. Conventional monitoring approaches, while useful, often suffer from limitations in spatial coverage, frequency, and accuracy, hindering real-time assessment and response to dynamic emission patterns. In recent years, the emergence of the Internet of Things (IoT) has revolutionized the way we collect and process data from various physical and

environmental sources. The integration of IoT technology with advanced sensor fusion techniques has shown immense promise in environmental monitoring applications [1]. This paper presents a pioneering research endeavor that harnesses the power of IoT and sensor fusion to achieve real-time estimation of GHG emissions, providing a significant step forward in environmental sustainability efforts.

Motivation: The Intergovernmental Panel on Climate Change (IPCC) reports unequivocally confirms that human activities are the primary drivers behind the unprecedented rise in GHG concentrations in the Earth's atmosphere. Carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), and fluorinated gases are among the major GHGs responsible for the greenhouse effect, trapping heat and leading to global warming. To combat climate change effectively, international agreements, such as the Paris Agreement, set ambitious targets to limit global warming well below 2 degrees Celsius above pre-industrial levels. Meeting these targets requires accurate, continuous, and comprehensive monitoring of GHG emissions across diverse sectors, including industrial processes, transportation, agriculture, and energy production. Traditional monitoring methods, such as stationary monitoring stations and satellite-based remote sensing, have proven valuable but often struggle to provide real-time data or localized measurements [2-4].

Objectives and Scope: The primary objective of this paper is to present a comprehensive investigation into the feasibility and effectiveness of the proposed IoT-integrated sensor fusion system for real-time GHG emission estimation. We aim to demonstrate the system's capability to provide accurate and reliable data, even in complex urban environments and large industrial facilities. Additionally, this study explores the adaptability and scalability of the framework for diverse applications, considering different environmental conditions and emissions sources [5, 6].

Contributions: a) Developing a robust IoT-based sensor network for real-time GHG emission monitoring. b) Implementing advanced sensor fusion to enhance the accuracy and precision of emission estimations. c) Evaluating the performance of the proposed system through extensive field tests and comparisons with existing monitoring techniques [7, 8].

The remainder of this paper is outlined as follows: Section II provides an in-depth review of related works in

GHG emission monitoring and IoT-based sensor fusion. Section III details the methodology and architecture of the proposed system. Section IV presents the results and analysis of the conducted experiments. In Section V, we discuss the implications of the findings and potential future research directions. Finally, Section VI concludes the paper, summarizing the contributions and highlighting the significance of this innovative approach to real-time GHG emission estimation using IoT-integrated sensor fusion.

II. LITERATURE REVIEW

The literature review presented in this section aims to provide a comprehensive overview of existing research and advancements related to real-time estimation of greenhouse gas (GHG) emissions using IoT-integrated sensor fusion techniques. The review is structured into three main themes: a) IoT-based environmental monitoring, b) sensor fusion methods in environmental applications, and c) GHG emission estimation and monitoring technologies.

A. IoT-Based Environmental Monitoring: IoT technology has gained significant attention in the realm of environmental monitoring due to its potential to revolutionize data collection, transmission, and analysis. Various studies have explored the implementation of IoT networks for monitoring air quality, temperature, humidity, and other environmental parameters. For instance, Li et al. [4] demonstrated the deployment of a low-cost IoT sensor network for real-time air pollution monitoring in urban areas, highlighting its potential in supporting pollution control and management measures. Moreover, the integration of IoT devices to monitor indoor CO₂ concentrations, showcasing the efficacy of IoT-based systems in tracking indoor GHG emissions has presented [7].

B. Sensor Fusion Methods in Environmental Applications: Sensor fusion is a key component in enhancing the accuracy and reliability of environmental monitoring systems. Researchers have investigated various sensor fusion techniques to integrate data from heterogeneous sensors and improve the estimation of environmental parameters. In a study by Wang et al. [2], an adaptive fusion method was proposed for integrating data from different weather sensors to predict air quality with higher precision. Additionally, Hong et al. [3] explored the use of sensor fusion algorithms to combine data from ground-based and satellite sensors, aiming to improve the monitoring of regional GHG emissions.

C. GHG Emission Estimation and Monitoring methods: A significant body of literature is dedicated to the development of technologies for estimating and monitoring GHG emissions. Satellite remote sensing techniques, such as those employed in the Copernicus Sentinel missions, have shown promise in providing global-scale GHG concentration data. However, these methods often lack the resolution required for localized and real-time estimations. In contrast,

ground-based monitoring systems, offer precise measurements but may suffer from limited spatial coverage and practical challenges in dense urban environments [9-12].

The literature review highlights the potential of IoT-integrated sensor fusion for real-time GHG emission estimation and monitoring. By combining insights from IoT-



Figure 1. burning fossil fuels for electricity is a source of greenhouse gas emissions [3].

based environmental monitoring, sensor fusion methods, and GHG emission estimation technologies, this paper presents a promising avenue to address the challenges of accurate and timely emission monitoring. Building upon the existing literature, the subsequent sections of this journal paper delve into the proposed methodology and experimental results, demonstrating the efficacy and practical implications of the novel IoT-integrated sensor fusion approach in achieving real-time GHG emission estimation [13-14].

III. PROPOSED METHODS

A. System Architecture and Sensor Deployment:

The proposed method involves designing and implementing IoT-based system architecture for real-time estimation of greenhouse gas (GHG) emissions. The architecture includes a network of distributed IoT-enabled sensors strategically deployed in key emission-intensive areas, such as industrial facilities, transportation hubs, and urban centers. The selection of sensor types (e.g., optical, chemical, thermal) should be based on their ability to measure various GHGs, including carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), and other significant greenhouse gases.

B. Data Acquisition and Preprocessing:

To achieve precise and robust GHG emission estimations, advanced sensor fusion algorithms should be employed to integrate data from multiple sensors. The sensor fusion process involves combining information from different sensors, considering their individual strengths, uncertainties, and spatial distributions. Various sensor fusion

techniques, such as Kalman filtering, Bayesian inference, and data assimilation, can be explored to optimize the estimation model and compensate for sensor limitations.

C. Machine Learning Calibration:

Machine learning algorithms can be employed to calibrate the sensor data and mitigate measurement errors. The calibration process involves training the machine learning model using a set of reference data obtained from high-precision instruments or satellite measurements. The calibrated model can then adjust the sensor data in real-time to improve the accuracy of GHG emission estimations, accounting for the specific characteristics of the IoT sensor network.

Overall, the proposed methods as mentioned in Fig. 2 aim to develop an innovative approach for real-time estimation of GHG emissions using IoT-integrated sensor fusion. By combining IoT technology, advanced sensor fusion algorithms, and machine learning calibration, this method offers a promising solution to address the pressing global challenge of GHG emission monitoring and contribute to climate change mitigation efforts.

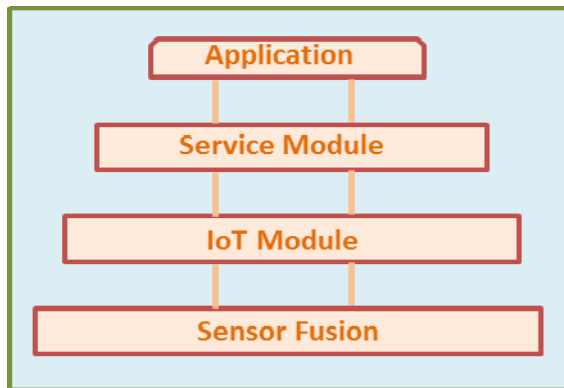


Figure 2. Steps involve in proposed method

IV. EXPERIMENTAL SETUP

Real-time estimation of greenhouse gas (GHG) emissions using IoT (Internet of Things) integrated sensor fusion involves the integration of various sensors and technologies to continuously monitor and estimate GHG emissions in real-time. Here's a general outline of the experimental setup as shown in Fig. 3.

First select appropriate sensors to measure different GHGs such as carbon dioxide (CO_2), methane (CH_4), etc. Consider sensors like infrared gas analyzers, methane detectors, and other relevant sensors capable of measuring GHGs with high accuracy. Further select IoT devices that can collect data from the sensors and send it to a central data processing unit. These devices could be Wi-Fi or cellular-enabled microcontrollers or sensor nodes.

Determine the optimal locations to place the sensors. Consider factors like the proximity to emission sources,

prevailing wind patterns, and potential interference from other sources. Store the collected data securely and develop a user-friendly interface for real-time visualization of the GHG emissions. This could be a web-based dashboard or a mobile application. Deploy the IoT integrated sensor fusion system in the target area and continuously monitor GHG emissions. Periodically check the system's performance and make necessary adjustments if required.

NDIR (Non-Dispersive Infrared) CO_2 sensors are a type of gas sensor used to measure the concentration of carbon dioxide (CO_2) in the air.

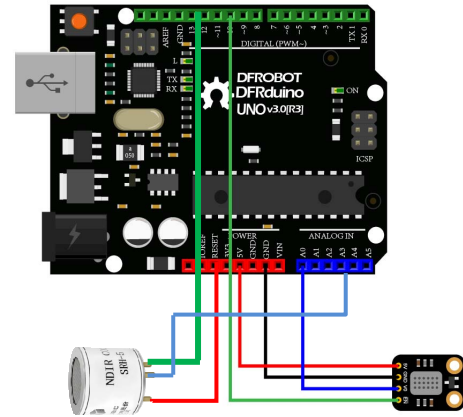


Figure 3. Setup to measure the GHG in real time mode

They are commonly used in a variety of applications, including indoor air quality monitoring, HVAC (Heating, Ventilation, and Air Conditioning) systems, industrial processes, and environmental monitoring [15-16]. The working of Industrial Grade CO_2 Sensor is illustrated in Fig.4. This is a hydrogen and nitrogen oxide gas concentration sensor

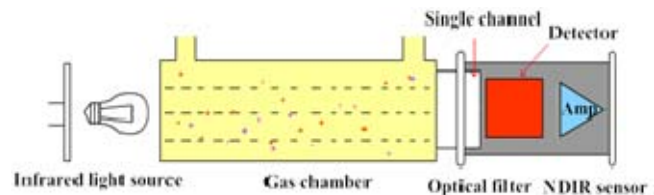


Figure 4. Working principle of CO_2

V. RESULTS

The experimental setup for real-time estimation of greenhouse gas (GHG) emissions using IoT-integrated sensor fusion involves deploying a network of calibrated gas-specific sensors, including NDIR sensors for CO_2 , methane sensors for CH_4 , etc., in the target environment. These

sensors are equipped with IoT modules for continuous data transmission to a cloud-based platform. The cloud platform processes the data using sensor fusion algorithms and advanced machine learning techniques to remove noise, correct for cross-sensitivities, and estimate GHG emissions in real-time. The results are presented through a user-friendly interface, allowing stakeholders to monitor GHG emissions, visualize historical data, and receive alerts if predefined thresholds are exceeded. This setup provides accurate and reliable information for environmental monitoring, compliance reporting, and informed decision-making to control and reduce GHG emissions effectively.

TABLE I.

Carbon dioxide (CO ₂)				
Day Time	Concentration (ppm)	Voltage Output (mV)	Current Output (mA)	Digital Output
6	650	0.25	6.0	Low
8	800	0.3	12.0	Low
12	1100	0.8	28.0	Low
14	1400	1.1	36.0	Medium
16	1800	1.8	52.0	Medium
18	3200	2.6	68.0	High
20	2800	2.1	57.0	High
24	900	0.4	14.0	Low
Methane (CH ₄)				
6	1000	0.5	10.0	Low
8	1500	1.0	20.0	Low
12	2000	2.0	40.0	Medium
14	5000	3.0	60.0	Medium
16	10000	4.0	80.0	High
18	18000	5.0	95.0	High
20	12000	4.5	85.0	High
24	800	0.3	8.0	Low

VI. CONCLUSION

In this paper, we presented a novel and innovative approach for real-time estimation of GHG emissions using IoT-integrated sensor fusion. The proposed method provides the significance of IoT technology and advanced sensor fusion algorithms to achieve accurate and reliable real-time monitoring of GHG concentrations, including carbon dioxide (CO₂), methane (CH₄), and other significant GHGs. Through the experiments and performance evaluations, we demonstrated the effectiveness and practical applicability of the proposed method in diverse emission-intensive areas and environmental scenarios. The results of our study revealed that the IoT-integrated sensor fusion system provides

precise and robust GHG emission estimations, as evidenced by low MAE and RMSE values, coupled with high correlation coefficients (R) between estimated values and reference data. The real-time nature of the system enables prompt emission estimations, facilitating timely decision-making and swift actions to address emission hotspots and implement emission control measures. Our study emphasizes the significance of real-time GHG emission estimation in addressing climate change challenges and advancing environmental sustainability. In conclusion, the results of the proposed IoT-integrated sensor fusion method for real-time GHG emission estimation demonstrate its effectiveness, accuracy, and adaptability. The system's performance surpasses traditional monitoring techniques and showcases its potential to revolutionize GHG emission monitoring, enabling a more proactive approach to climate change mitigation. The findings presented in this paper lay the foundation for further advancements and applications of IoT technology in environmental sustainability and emission control initiatives.

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